

* Unit - 4 *

Turing Machine

* Definition - A Turing machine (TM) is a mathematical model of computation.

Invented by Alan Turing in 1936.

- It is an accepting device.
- It accepts Recursively Enumerable Language.
- It has external memory which remember arbitrarily long sequence of input. (Left to right read on tape)

Importance - Most powerful machine.

- Solve complex problem
- Simulate all other machine (FA, PDA etc.)
- Represent real computers: conceptually.

Key Features

- (1) Infinite memory - works like memory
Can store unlimited data
- (2) Tape with cells - Each cell stores one symbol.
Blank symbol = B
- (3) Read/Write Head - On Tape.
(It can move left (L) or right (R) or stay.)
- (4) Input = Output Concept
TM can modify its input.
Same tape used for input & output.

Working

- (i) Read symbol from tape
- (ii) Apply transition
- (iii) Write new symbol
- (iv) Move head (L/R)
- (v) change state.

* Formal Definition of Turing Machine:-

A Turing Machine (TM) is defined as a 7-tuple.

$$M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$$

where Q : finite set of states = $\{q_0, q_1, q_2\}$
 Γ : finite set of tape symbol = $\{0, 1, X, B\}$
 Σ : finite set of inputs = $\{0, 1\}$
 δ : Transition states =
 q_0 : Start State $q_0 \in Q$
 B : $B \in \Gamma$ blank symbol
 F : Set of Final state.

$$\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$$

Ex If TM is in state q_i & reads a symbol, then

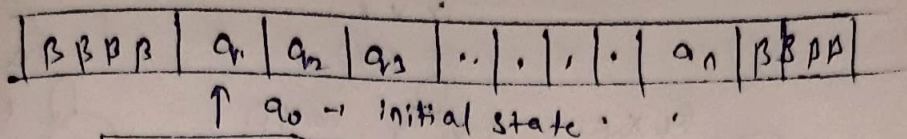
$$\delta(q_i, a) = (q_j, b, R)$$

q_j → change state

$a \rightarrow b$ replace

more right.

Note If TM halts in a final state → Accept
otherwise → Reject or Loop.



Read / write
head

Basic model of TM.

A configuration shows current state, head position & Tape content

* TM as Language Acceptors :-

It checks whether a given string belongs to a language or not

$\Rightarrow$ A TM M accepts a language L if:

$$L = \{ w \mid M \text{ accepts } w \}$$

Three possible outcomes -

(i) ACCEPT :- TM enters a final state (q_{GF})

String belongs to language.

Acceptance by Final state

(ii) REJECT :- TM halts in a non-final state

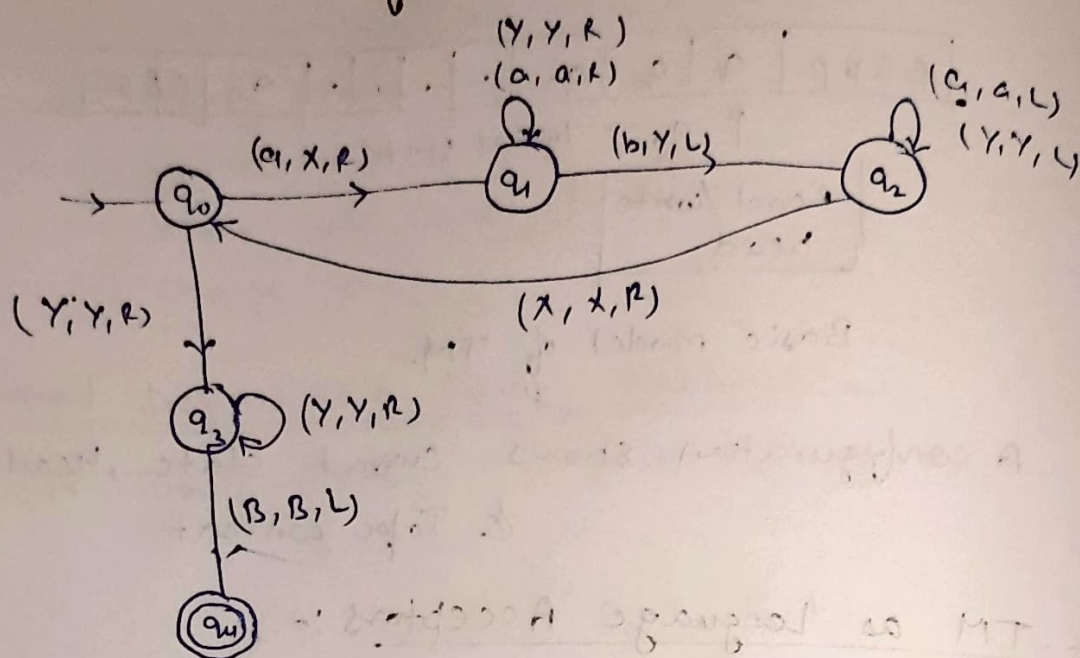
String Not in language

Acceptance by Halting.

(iii) LOOP :- TM never stops

Infinite computation.

Ex Design a TM for $\{a^n b^n \mid n \geq 1\}$



Ex Design a TM for $\{a^n b^n c^n \mid n \geq 1\}$

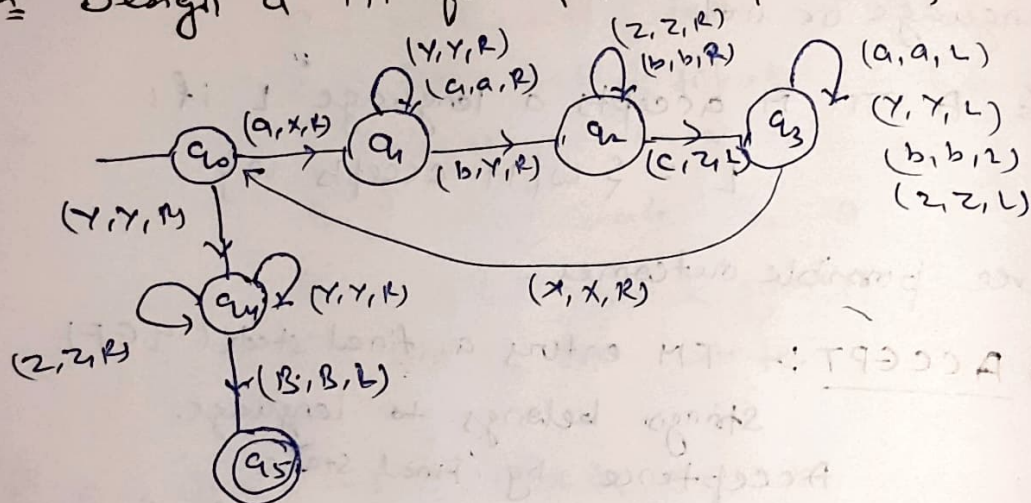


Table.

State	a	b	c	x	y	z	B
q ₀	(q ₁ , x, R)	-	-	-	(q ₄ , y, R)	-	-
q ₁	(q ₁ , a, R)	(q ₃ , y, R)	-	-	(q ₁ , y, R)	-	-
q ₂	-	(q ₂ , b, R)	(q ₃ , z, L)	(q ₀ , x, R)	-	(q ₂ , z, R)	-
q ₃	(q ₃ , a, L)	(q ₃ , b, L)	-	(q ₀ , x, R)	(q ₃ , y, L)	(q ₃ , z, L)	-
q ₄	-	-	-	-	(q ₄ , y, R)	(q ₄ , z, R)	(q ₅ , B, L)
q ₅	-	-	-	-	-	-	-

* Turing Machine as Transducers:-

1) Transducer machine that takes input & produces output.

When a TM is used to

- Transform input string into another string
- It is called Transducer.

Note Acceptor

output - Yes/No
check language

$$G_x \quad f(w) = \text{output}$$

input a a a

Transducer

output - Modified string.

Check computation function

TM computes a function.

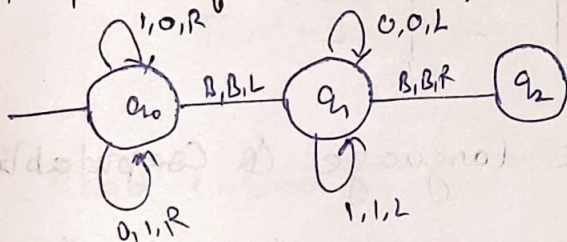
TM converts bbb

Types of computation

$\Rightarrow$ Recursive language (RL)

- String transformation
- Computation
- Function evaluation.

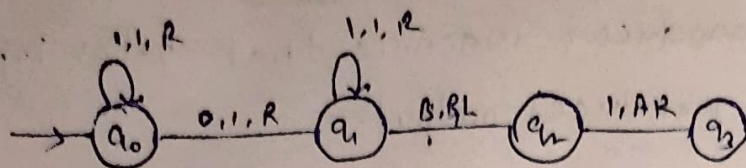
Design T.M. for 1's complement



Transition table

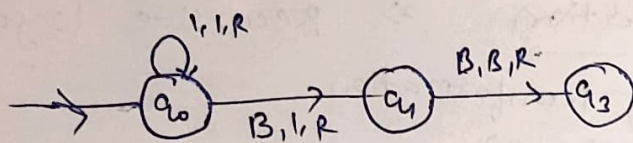
States	0	1	B
q_0	1 R q_0	0 L q_1	B L q_1
q_1	0 L q_1	1 R q_2	B R q_2
q_2	-	-	-

Q Design a TM to find sum of 2 unary number.



State	0	1	B
q_0	1R q_1	1R q_0	-
q_1	-	1R q_1	BL q_2
q_2	-	BRA q_3	-
q_3	-	-	-

Q Design a TM to increment a unary number
i.e. $f(n) = n + 1$



State	1	B
q_0	1R q_0	BR q_1
q_1	-	BR q_2
q_2	-	-

Note Computable language & Computable Functions

(i) Computable language

(ii) Non-Computable language

- Check equality
- always halt
- 1 is computable

if there exists a TM
that decides it.

- Some languages
No TM can decide them
- Ex: Halting problem.

Computable function

A function f is computable if there exists a TM that computes it.

$$\underline{G3} \quad f: \Sigma^* \rightarrow \Sigma^*$$

That means

- TM halts
- Produces output $f(w)$

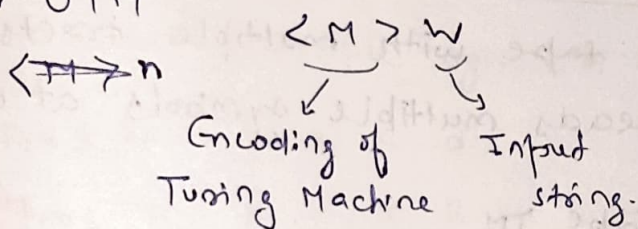
*Universal Turing Machine:-

A Turing Machine

A universal Turing machine (UTM) is a special TM that : can simulate any other Turing Machine.

- Instead of designing separate TM for each problem one machine (UTM) can run all TMs.

Input for UTM



Working

- Read encoded TM ($\langle M \rangle$)
- Read input string (w)
- Simulate M on w

UTM gives same results as TM.

Accept / Reject or Loop.

UTM is usually designed using Multi-tape Turing Machine.

* Modification of Turing Machine

Many variations of TM exist but all have same power.

Types of modification

1.) TM with stay option

Head can - Move Left (L)

Move Right (R)

Stay

No change in power.

2.) Semi-infinite Tape

Tape infinite in only one direction.

Still same power.

3.) Multi Track TM

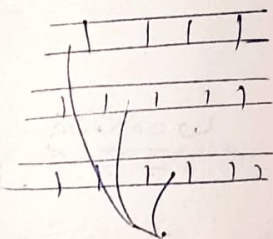
Single tape with multiple tracks.

Head reads multiple symbols at once.

4.) Multi Tape TM

Multiple tape + heads

Faster but same power.



5.) Multi Dimensional TM

Tape in 2D or more dimensions

Moves left, right, up & down.

6.) Jumping TM

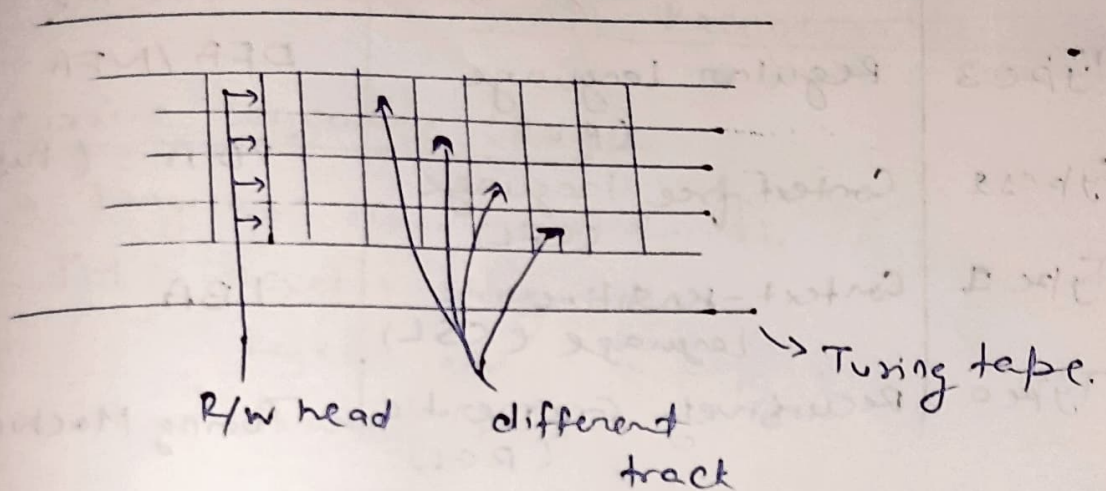
Head can jump multiple cells

Still same power

- 7) Non-erasing TM
 cannot erase symbols
 still equivalent

* Multiple Tracks Turing Machine

A multiple track TM is a TM with single tape.
 But divided into multiple tracks (rows).



Each cell stores multiple symbols:

E_1 Track 1 : a b c d
 Track 2 : x y z B

Only one head with reads all track simultaneously

Transition function

$$\delta(q, (a_1, a_2, a_3 \dots a_k)) = (q', (b_1, b_2 \dots b_k), D)$$

Need

Make design easier

- Instead of modifying input repeatedly
- Use extra track for memory

It increases efficiency.

* Hierarchy of Formal Language:-

Arrangement of Languages based on power of machines.

• Chomsky Hierarchy,

There are 4 types of language

Type	Language	Machine
Type 3	Regular Language (RL)	DFA / NFA
Type 2	Context free language (CFL)	PDA (Pushdown Automata)
Type 1	Context-sensitive Language (CSL)	LBA
Type 0	Recursively Enumerable (REL)	Turing Machine

• Inclusion Relation

$RL \subset CFL \subset CSL \subset REL$

Means

every RL is CFL

every CFL is CSL

every CSL is REL

REL (most powerful)

↓
CSL

↓
CFL

↓
RL

(Least powerful)

* Recursively Enumerable language & Recursive:-

A language L is REL if

There exists a TM that:

- Accept all strings in L

- May loop for strings not in L .

* TM may not Halt.

This is also called "Turing Recognizable Language."

* Recursive Language (RL):-

A language is recursive if:

TM: accepts strings in L

Rejects strings not in L

Always Halts.

→ It always gives answer in YES / NO.

Relationship

$$RL \subset REL$$

means every recursive language is REL

But not every REL is recursive.

Also known as "Turing decidable language".

* Recognizer vs Decider:-

Recognizer

TM that Accepts strings in L

- May loop for others

used for REL.

Decider

TM that Accepts or rejects

- Always halts.

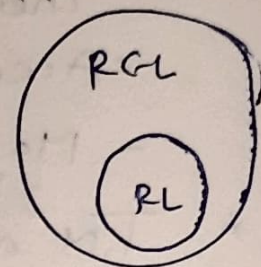
Used for Recursive language

* Properties of RL & RGL ... (Recursive lang.)

(1) Properties of Recursive language)

Closed under

- Union ($L_1 \cup L_2$)
- Intersection ($L_1 \cap L_2$)
- Complement ($\Sigma^* - L_1$)
- Concatenation ($L_1 \cdot L_2$)
- Kleen star (L_1^*)



(2) Properties of Recursive enumerable language

Closed under

- Union
- Concatenation
- Kleen star
- Intersection

Not closed under

- Complement
- Difference

* Context Sensitive Grammar :- (CFG)

A grammar is context sensitive if

$$\alpha A \beta \rightarrow \alpha \gamma \beta \quad \beta, \alpha \in (\Sigma \cup V_n)^*$$

where γ is not empty. $\gamma \beta \in (\Sigma \cup V_n)^+$

Replacement depends on context. $\cancel{V_n \in (\Sigma \cup V_n)^*}$, $A \in V_n$

→ Special Rule

$S \rightarrow \epsilon$ allowed only if S not on RHS.

$$\cancel{\alpha} \leq \beta$$

Power more powerful than CFG.

* Context Sensitive Language (CSL)

Language generated by ~~CFG~~ CSG.

* Accepted by Linear Bounded Automata (LBA)

$$\underline{\text{Ex}} \quad L = \{a^n b^n c^n\}$$

not CFL

but CSL

$$S \rightarrow aTb \mid cb$$

$$aT \rightarrow aaTb \mid ac$$

properties

closed under:-

- Union
- Intersection
- Complement
- Concatenation
- Kleen star
- Reversal

$$L(n) = \{ab\} \cup \{a^n c b^n \mid n > 0\}$$